HealthTrack Conceptual Data Model

Entities and relationships, read left to right along each line. Attributes, keys, and types are on the next page.

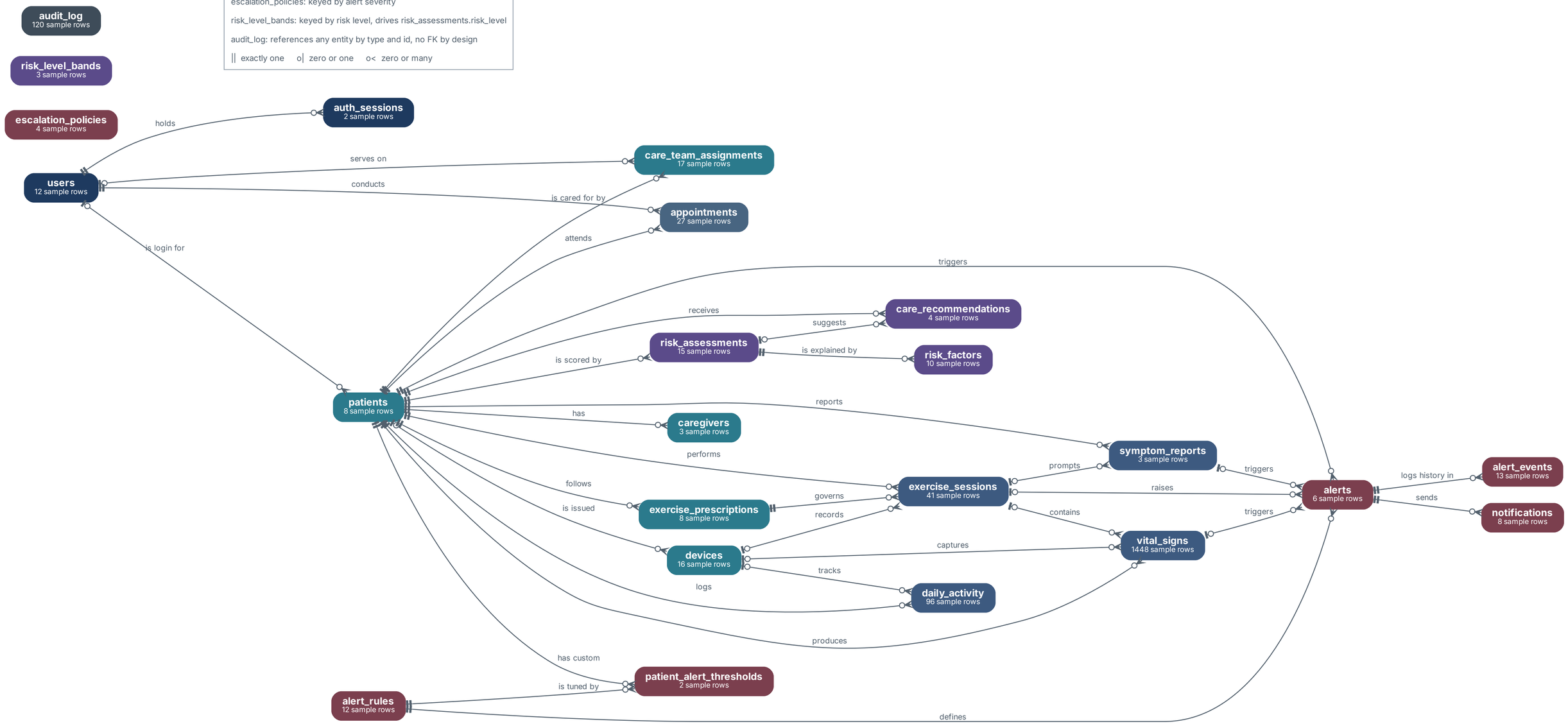
Stand-alone configuration and audit

escalation_policies: keyed by alert severity

risk_level_bands: keyed by risk level, drives risk_assessments.risk_level

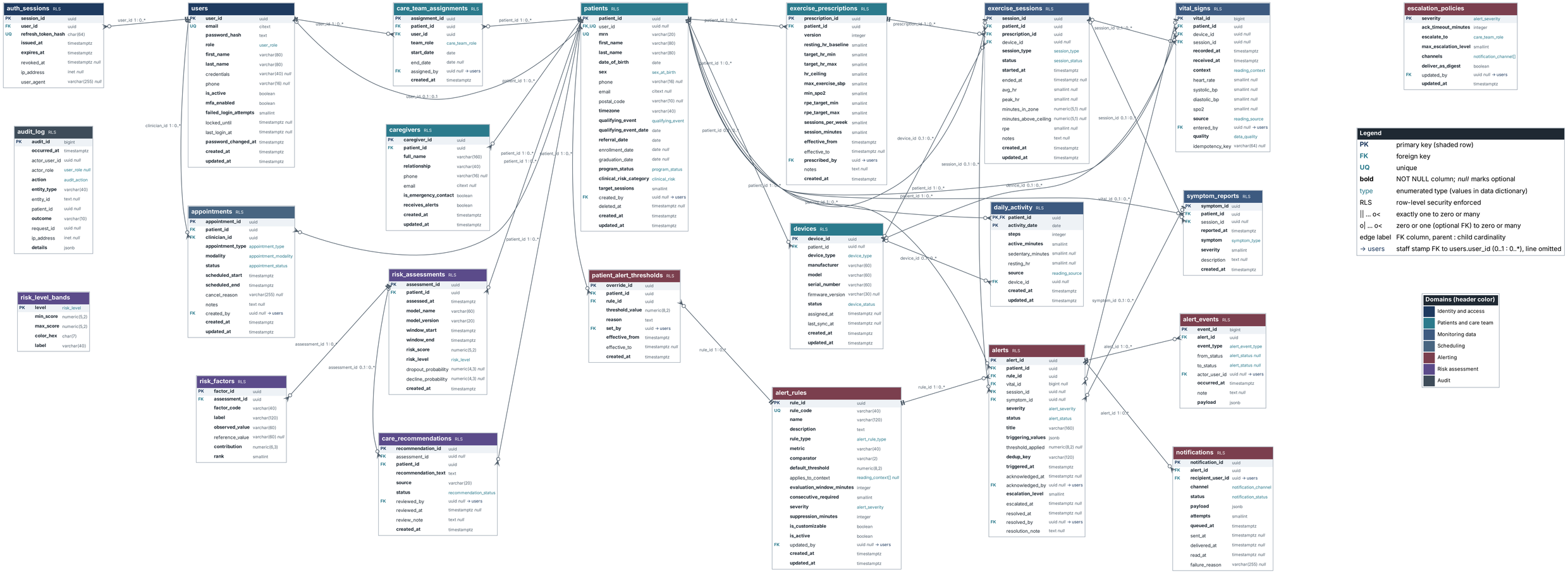
audit_log: references any entity by type and id, no FK by design

|| exactly one o| zero or one o< zero or many



HealthTrack Entity-Relationship Diagram

PostgreSQL schema healthtrack: 23 tables, crow's foot notation, generated from the live schema catalog



HealthTrack Data Dictionary and Validation Rules

Generated from the live PostgreSQL catalog after running 01 to 04. Schema `healthtrack`, 23 tables, 26 enumerated types, 5 views. Every rule below is enforced in the database and exercised by `05_validation_tests.sql`.

Domain-specific validation rules

Vital signs	Enforced by
Reject physiologically impossible values: heart rate 20 to 300, systolic 50 to 300, diastolic 20 to 200, SpO2 50 to 100	`CHECK`
Flag unlikely but possible values as suspect and exclude them from alerting: heart rate outside 30 to 220, systolic outside 70 to 250, diastolic outside 35 to 140, SpO2 below 70, pulse pressure under 15	`trigger vitals_plausibility`
Systolic must exceed diastolic, and the two are stored as a pair	`CHECK`
Every reading carries at least one measurement	`CHECK`
Manual readings name the person who entered them; device readings name the device	`CHECK`
No timestamps in the future (5 minute clock-skew allowance)	`trigger`
Device uploads are idempotent: a replayed batch cannot create duplicates	`UNIQUE (patient_id, idempotency_key)`

Patients and care team	Enforced by
Patient is at least 18 at the qualifying event; event, referral, enrollment, and graduation dates are in order	`CHECK`
Active, paused, and graduated patients have an enrollment date; graduated status and graduation date go together	`CHECK`
An active home-program patient must be low or moderate clinical risk (AACVPR stratification)	`CHECK`
MRN, E.164 phone, ZIP code, and email formats	`CHECK`
A patient's login account must hold the patient role	`trigger validate_roles`
Exactly one current primary clinician and one current supervising physician per patient	`partial UNIQUE index`
Supervising physician must be a physician; care team members must be clinicians or physicians	`trigger validate_roles`

Exercise prescription and sessions	Enforced by
Resting baseline < zone minimum < zone maximum < heart rate ceiling; zone minimum at least 40, ceiling at most 200	`CHECK`
RPE targets on the Borg 6 to 20 scale, minimum not above maximum	`CHECK`
Prescriptions are signed by a physician	`trigger validate_roles`
One current prescription per patient; a new version numbers itself and closes the previous one	`partial UNIQUE + trigger rx_versioning`
Session ends after it starts, lasts at most 4 hours, and closed sessions have an end time	`CHECK`
Peak heart rate at least the average; minutes in zone cannot exceed session length; RPE 6 to 20	`CHECK`
A patient cannot be in two overlapping sessions	`EXCLUDE (gist)`

Daily steps 0 to 100,000; active plus sedentary minutes at most 1,440; one row per patient per day	`CHECK + PK`
Symptom severity 0 to 10; a symptom of type other requires a description	`CHECK`

Appointments	Enforced by
Length between 10 minutes and 3 hours; cancellation requires a reason	`CHECK`
No double-booking of a clinician or a patient	`EXCLUDE (gist)`
At most one graduation assessment per patient	`partial UNIQUE index`

Alerting	Enforced by
Critical rules cannot have a suppression window	`CHECK`
One open alert per patient, rule, and underlying condition (duplicate suppression)	`partial UNIQUE (dedup_key)`
Acknowledgment time and user are set together and not before the trigger time; resolution requires resolver, time, and note	`CHECK`
Only clinicians and physicians can acknowledge or resolve alerts	`trigger validate_roles`
Critical alerts escalate within 30 minutes and are never batched; only low alerts may go to the daily digest	`CHECK`
Notification payload must contain alert_id, patient_id, severity, rule_code, and triggered_at	`CHECK (jsonb ?&)`
Patient-specific thresholds only on customizable rules, set by a physician, one current override per rule	`trigger + partial UNIQUE`
Every alert creation and status change is recorded in alert_events, which cannot be edited	`trigger alert_history + append-only`

Risk and audit	Enforced by
Risk score 0 to 100 and probabilities 0 to 1; model version is semantic (x.y.z)	`CHECK`
Risk level is always derived from the configured, non-overlapping score bands	`trigger + EXCLUDE`
System-generated recommendations link to their assessment; approval or rejection requires a clinician or physician reviewer	`CHECK + trigger`
Audit log cannot be updated, deleted, or truncated	`trigger forbid_change`
Every PHI write is audited automatically by column name, without copying PHI values	`trigger audit_phi_change`

Table definitions

users 12 sample rows

Login accounts for patients and staff; role drives API authorization and row-level security.

Column	Type	Null	Key	Default	Notes
user_id	uuid		PK	gen_random_uuid()	
email	citext		UQ		
password_hash	text				
role	user_role				values: patient, clinician, physician, admin, it_admin
first_name	varchar(80)				
last_name	varchar(80)				
credentials	varchar(40)	yes			
phone	varchar(16)	yes			
is_active	boolean			true	
mfa_enabled	boolean			false	
failed_login_attempts	smallint			0	
locked_until	timestamptz	yes			
last_login_at	timestamptz	yes			
password_changed_at	timestamptz			now()	
created_at	timestamptz			now()	
updated_at	timestamptz			now()	

Constraint, index, or trigger	Definition
users_email_check	CHECK ((email ~ '^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$'))
users_failed_login_attempts_check	CHECK (((failed_login_attempts >= 0) AND (failed_login_attempts <= 100)))
users_first_name_check	CHECK ((btrim((first_name)) <> ''))
users_last_name_check	CHECK ((btrim((last_name)) <> ''))
users_password_hash_check	CHECK ((password_hash ~ '^\$(2[aby]\$[0-9]{2}\$|argon2id\$)'))
users_phone_check	CHECK (((phone) ~ '^\+[1-9][0-9]{7,14}$'))
users_email_key	UNIQUE (email)

auth_sessions 2 sample rows row-level security

Refresh-token sessions; only a SHA-256 hash of each token is stored, so a database leak does not expose live tokens.

Column	Type	Null	Key	Default	Notes
session_id	uuid		**PK**	gen_random_uuid()	
user_id	uuid		**FK users.user_id**		
refresh_token_hash	char(64)		**UQ**		
issued_at	timestamptz			now()	
expires_at	timestamptz				
revoked_at	timestamptz	yes			
ip_address	inet	yes			
user_agent	varchar(255)	yes			

Constraint, index, or trigger	Definition
auth_sessions_expiry_ck	CHECK (((expires_at > issued_at) AND (expires_at <= (issued_at + '7 days'))))
auth_sessions_refresh_token_hash_check	CHECK ((refresh_token_hash ~ '^[0-9a-f]{64}$'))
auth_sessions_revoked_ck	CHECK (((revoked_at IS NULL) OR (revoked_at >= issued_at)))
auth_sessions_refresh_token_hash_key	UNIQUE (refresh_token_hash)
auth_sessions_user_ix	INDEX auth_sessions_user_ix (user_id) WHERE (revoked_at IS NULL)

patients 8 sample rows row-level security

Rehab patients from referral to graduation. Soft-deleted, never hard-deleted.

Column	Type	Null	Key	Default	Notes
patient_id	uuid		**PK**	gen_random_uuid()	
user_id	uuid	yes	**FK users.user_id, UQ**		
mrn	varchar(20)		**UQ**		
first_name	varchar(80)				
last_name	varchar(80)				
date_of_birth	date				
sex	sex_at_birth				values: female, male, intersex, unknown
phone	varchar(16)	yes			
email	citext	yes			

postal_code	varchar(10)	yes			
timezone	varchar(40)			'America/Denver'	
qualifying_event	qualifying_event				values: myocardial_infarction, pci, cabg, valve_surgery, stable_angina, heart_failure_reduced_ef, heart_transplant
qualifying_event_date	date				
referral_date	date				
enrollment_date	date	yes			
graduation_date	date	yes			
program_status	program_status			'referred'	values: referred, active, paused, graduated, withdrawn
clinical_risk_category	clinical_risk				AACVPR exercise risk stratification at intake. values: low, moderate, high
target_sessions	smallint			36	
created_by	uuid	yes	**FK users.user_id**		
deleted_at	timestamptz	yes			
created_at	timestamptz			now()	
updated_at	timestamptz			now()	

Constraint, index, or trigger	Definition
patients_adult_ck	CHECK ((date_of_birth <= (qualifying_event_date - '18 years')))
patients_date_of_birth_check	CHECK ((date_of_birth > '1900-01-01'))
patients_email_check	CHECK ((email ~ '^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$'))
patients_enroll_order_ck	CHECK (((enrollment_date IS NULL) OR (enrollment_date >= referral_date)))
patients_enrolled_ck	CHECK (((program_status = ANY (ARRAY['referred', 'withdrawn'])) OR (enrollment_date IS NOT NULL)))
patients_event_order_ck	CHECK ((qualifying_event_date <= referral_date))
patients_first_name_check	CHECK ((btrim((first_name)) <> ''))
patients_grad_order_ck	CHECK (((graduation_date IS NULL) OR (graduation_date >= enrollment_date)))
patients_graduated_ck	CHECK (((program_status = 'graduated') = (graduation_date IS NOT NULL)))
patients_home_risk_ck	CHECK (((program_status <> 'active') OR (clinical_risk_category = ANY (ARRAY['low', 'moderate']))))
patients_last_name_check	CHECK ((btrim((last_name)) <> ''))
patients_mrn_check	CHECK (((mrn) ~ '^[A-Z0-9-]{6,20}$'))
patients_phone_check	CHECK (((phone) ~ '^\+[1-9][0-9]{7,14}$'))
patients_postal_code_check	CHECK (((postal_code) ~ '^[0-9]{5}(-[0-9]{4})?$'))

patients_target_sessions_check	CHECK (((target_sessions >= 1) AND (target_sessions <= 72)))
patients_mrn_key	UNIQUE (mrn)
patients_user_id_key	UNIQUE (user_id)
patients_status_ix	INDEX patients_status_ix (program_status) WHERE (deleted_at IS NULL)
patients_audit_trg	trigger function audit_phi_change()
patients_roles_trg	trigger function validate_roles()

caregivers

3 sample rows · row-level security

Family or emergency contacts; may opt in to alert notifications.

Column	Type	Null	Key	Default	Notes
caregiver_id	uuid		**PK**	gen_random_uuid()	
patient_id	uuid		**FK patients.patient_id**		
full_name	varchar(160)				
relationship	varchar(40)				
phone	varchar(16)	yes			
email	citext	yes			
is_emergency_contact	boolean			false	
receives_alerts	boolean			false	
created_at	timestamptz			now()	
updated_at	timestamptz			now()	

Constraint, index, or trigger	Definition
caregivers_contact_ck	CHECK (((phone IS NOT NULL) OR (email IS NOT NULL)))
caregivers_email_check	CHECK ((email ~ '^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$'))
caregivers_full_name_check	CHECK ((btrim((full_name)) <> ''))
caregivers_phone_check	CHECK (((phone) ~ '^\+[1-9][0-9]{7,14}$'))
caregivers_audit_trg	trigger function audit_phi_change()

care_team_assignments

17 sample rows · row-level security

Who is responsible for each patient and when; drives alert routing and row-level access.

Column	Type	Null	Key	Default	Notes

assignment_id	uuid		**PK**	gen_random_uuid()	
patient_id	uuid		**FK patients.patient_id**		
user_id	uuid		**FK users.user_id**		
team_role	care_team_role				values: primary_clinician, supervising_physician, covering_clinician
start_date	date			CURRENT_DATE	
end_date	date	yes			
assigned_by	uuid	yes	**FK users.user_id**		
created_at	timestamptz			now()	

Constraint, index, or trigger	Definition
care_team_dates_ck	CHECK (((end_date IS NULL) OR (end_date >= start_date)))
care_team_one_primary_ux	UNIQUE INDEX care_team_one_primary_ux (patient_id, team_role) WHERE ((end_date IS NULL) AND (team_role = ANY (ARRAY['primary_clinician', 'supervising_physician'])))
care_team_user_ix	INDEX care_team_user_ix (user_id) WHERE (end_date IS NULL)
care_team_assignments_audit_trg	trigger function audit_phi_change()
care_team_roles_trg	trigger function validate_roles()

exercise_prescriptions

8 sample rows · row-level security

Versioned, physician-signed exercise prescription with heart rate zone and safety limits.

Column	Type	Null	Key	Default	Notes
prescription_id	uuid		**PK**	gen_random_uuid()	
patient_id	uuid		**FK patients.patient_id**		
version	integer				
resting_hr_baseline	smallint				
target_hr_min	smallint				
target_hr_max	smallint				
hr_ceiling	smallint				
max_exercise_sbp	smallint			200	
min_spo2	smallint			90	
rpe_target_min	smallint			11	

rpe_target_max	smallint			14	
sessions_per_week	smallint				
session_minutes	smallint				
effective_from	timestamptz			now()	
effective_to	timestamptz	yes			
prescribed_by	uuid		**FK users.user_id**		
notes	text	yes			
created_at	timestamptz			now()	

Constraint, index, or trigger	Definition
exercise_prescriptions_max_exercise_sbp_check	CHECK (((max_exercise_sbp >= 140) AND (max_exercise_sbp <= 250)))
exercise_prescriptions_min_spo2_check	CHECK (((min_spo2 >= 85) AND (min_spo2 <= 95)))
exercise_prescriptions_resting_hr_baseline_check	CHECK (((resting_hr_baseline >= 30) AND (resting_hr_baseline <= 120)))
exercise_prescriptions_session_minutes_check	CHECK (((session_minutes >= 10) AND (session_minutes <= 90)))
exercise_prescriptions_sessions_per_week_check	CHECK (((sessions_per_week >= 1) AND (sessions_per_week <= 7)))
exercise_prescriptions_version_check	CHECK ((version >= 1))
rx_hr_order_ck	CHECK (((resting_hr_baseline < target_hr_min) AND (target_hr_min < target_hr_max) AND (target_hr_max < hr_ceiling) AND (target_hr_min >= 40) AND (hr_ceiling <= 200)))
rx_period_ck	CHECK (((effective_to IS NULL) OR (effective_to > effective_from)))
rx_rpe_ck	CHECK ((((rpe_target_min >= 6) AND (rpe_target_min <= 20)) AND ((rpe_target_max >= 6) AND (rpe_target_max <= 20)) AND (rpe_target_min <= rpe_target_max)))
rx_version_uq	UNIQUE (patient_id, version)
rx_one_current_ux	UNIQUE INDEX rx_one_current_ux (patient_id) WHERE (effective_to IS NULL)
exercise_prescriptions_audit_trg	trigger function audit_phi_change()
rx_roles_trg	trigger function validate_roles()
rx_versioning_trg	trigger function rx_versioning()

devices

16 sample rows · row-level security

Issued home devices; unassigned inventory is managed by IT.

Column	Type	Null	Key	Default	Notes
device_id	uuid		**PK**	gen_random_uuid()	
patient_id	uuid	yes	**FK patients.patient_id**		

device_type	device_type				values: heart_rate_band, blood_pressure_cuff, pulse_oximeter, activity_tracker
manufacturer	varchar(60)				
model	varchar(60)				
serial_number	varchar(60)				
firmware_version	varchar(30)	yes			
status	device_status			'inactive'	values: active, inactive, lost, retired
assigned_at	timestamptz	yes			
last_sync_at	timestamptz	yes			
created_at	timestamptz			now()	
updated_at	timestamptz			now()	

Constraint, index, or trigger	Definition
devices_active_ck	CHECK (((status <> 'active') OR ((patient_id IS NOT NULL) AND (assigned_at IS NOT NULL))))
devices_serial_uq	UNIQUE (manufacturer, serial_number)
devices_patient_ix	INDEX devices_patient_ix (patient_id) WHERE (status = 'active')

exercise_sessions

41 sample rows · row-level security

One home exercise session, with metrics derived from its readings.

Column	Type	Null	Key	Default	Notes
session_id	uuid		**PK**	gen_random_uuid()	
patient_id	uuid		**FK patients.patient_id**		
prescription_id	uuid		**FK exercise_prescriptions.prescription_id**		
device_id	uuid	yes	**FK devices.device_id**		
session_type	session_type			'aerobic'	values: aerobic, resistance, combined, flexibility
status	session_status			'in_progress'	values: in_progress, completed, partial, aborted
started_at	timestamptz				

ended_at	timestamptz	yes		
avg_hr	smallint	yes		
peak_hr	smallint	yes		
minutes_in_zone	numeric(5,1)	yes		
minutes_above_ceiling	numeric(5,1)	yes		
rpe	smallint	yes		
notes	text	yes		
created_at	timestamptz		now()	
updated_at	timestamptz		now()	

Constraint, index, or trigger	Definition
exercise_sessions_avg_hr_check	CHECK (((avg_hr >= 30) AND (avg_hr <= 250)))
exercise_sessions_minutes_above_ceiling_check	CHECK ((minutes_above_ceiling >= (0)))
exercise_sessions_minutes_in_zone_check	CHECK ((minutes_in_zone >= (0)))
exercise_sessions_peak_hr_check	CHECK (((peak_hr >= 30) AND (peak_hr <= 250)))
exercise_sessions_rpe_check	CHECK (((rpe >= 6) AND (rpe <= 20)))
sessions_closed_ck	CHECK (((status = 'in_progress') OR (ended_at IS NOT NULL)))
sessions_peak_ck	CHECK (((peak_hr IS NULL) OR (avg_hr IS NULL) OR (peak_hr >= avg_hr)))
sessions_time_ck	CHECK (((ended_at IS NULL) OR ((ended_at > started_at) AND (ended_at <= (started_at + '04:00:00')))))
sessions_zone_len_ck	CHECK (((minutes_in_zone IS NULL) OR (ended_at IS NULL) OR (minutes_in_zone <= (EXTRACT(epoch FROM (ended_at - started_at)) / 60.0))))
sessions_no_overlap	EXCLUDE USING gist (patient_id WITH =, tstzrange(started_at, ended_at) WITH &&) WHERE ((ended_at IS NOT NULL))
exercise_sessions_audit_trg	trigger function audit_phi_change()

vital_signs

1448 sample rows · row-level security

Heart rate, blood pressure, and SpO2 readings from devices or manual entry.

Column	Type	Null	Key	Default	Notes
vital_id	bigint		**PK**	identity	
patient_id	uuid		**FK patients.patient_id**		
device_id	uuid	yes	**FK devices.device_id**		
session_id	uuid	yes	**FK exercise_sessions.session_id**		

recorded_at	timestamptz				
received_at	timestamptz			now()	
context	reading_context				values: rest, warmup, exercise, cooldown, recovery
heart_rate	smallint	yes			
systolic_bp	smallint	yes			
diastolic_bp	smallint	yes			
spo2	smallint	yes			
source	reading_source			'device'	values: device, manual
entered_by	uuid	yes	**FK users.user_id**		
quality	data_quality			'valid'	values: valid, suspect, artifact
idempotency_key	varchar(64)	yes			

Constraint, index, or trigger	Definition
vital_signs_diastolic_bp_check	CHECK (((diastolic_bp >= 20) AND (diastolic_bp <= 200)))
vital_signs_heart_rate_check	CHECK (((heart_rate >= 20) AND (heart_rate <= 300)))
vital_signs_idempotency_key_check	CHECK (((idempotency_key) ~ '^[A-Za-z0-9._:-]{8,64}$'))
vital_signs_spo2_check	CHECK (((spo2 >= 50) AND (spo2 <= 100)))
vital_signs_systolic_bp_check	CHECK (((systolic_bp >= 50) AND (systolic_bp <= 300)))
vitals_any_value_ck	CHECK ((num_nonnulls(heart_rate, systolic_bp, diastolic_bp, spo2) > 0))
vitals_bp_order_ck	CHECK (((systolic_bp IS NULL) OR (systolic_bp > diastolic_bp)))
vitals_bp_pair_ck	CHECK (((systolic_bp IS NULL) = (diastolic_bp IS NULL)))
vitals_device_ck	CHECK (((source <> 'device') OR (device_id IS NOT NULL)))
vitals_manual_ck	CHECK (((source <> 'manual') OR (entered_by IS NOT NULL)))
vitals_received_ck	CHECK ((received_at >= (recorded_at - '00:05:00')))
vitals_idempotency_uq	UNIQUE (patient_id, idempotency_key)
vitals_session_ix	INDEX vitals_session_ix (session_id) WHERE (session_id IS NOT NULL)
vital_signs_audit_trg	trigger function audit_phi_change()
vitals_plausibility_trg	trigger function vitals_plausibility()

daily_activity

96 sample rows · row-level security

One row per patient per day from the activity tracker.

Column	Type	Null	Key	Default	Notes
patient_id	uuid		**PK, FK patients.patient_id**		
activity_date	date		**PK**		
steps	integer				
active_minutes	smallint				
sedentary_minutes	smallint	yes			
resting_hr	smallint	yes			
source	reading_source			'device'	values: device, manual
device_id	uuid	yes	**FK devices.device_id**		
created_at	timestamptz			now()	
updated_at	timestamptz			now()	

Constraint, index, or trigger	Definition
activity_minutes_ck	CHECK (((active_minutes + COALESCE((sedentary_minutes), 0)) <= 1440))
daily_activity_active_minutes_check	CHECK (((active_minutes >= 0) AND (active_minutes <= 1440)))
daily_activity_resting_hr_check	CHECK (((resting_hr >= 25) AND (resting_hr <= 150)))
daily_activity_sedentary_minutes_check	CHECK (((sedentary_minutes >= 0) AND (sedentary_minutes <= 1440)))
daily_activity_steps_check	CHECK (((steps >= 0) AND (steps <= 100000)))
daily_activity_audit_trg	trigger function audit_phi_change()

symptom_reports

3 sample rows · row-level security

Patient-reported symptoms, often after a session.

Column	Type	Null	Key	Default	Notes
symptom_id	uuid		**PK**	gen_random_uuid()	
patient_id	uuid		**FK patients.patient_id**		
session_id	uuid	yes	**FK exercise_sessions.session_id**		
reported_at	timestamptz			now()	
symptom	symptom_type				values: chest_pain, shortness_of_breath, dizziness, palpitations, unusual_fatigue, leg_pain, nausea, other

severity	smallint				
description	text	yes			
created_at	timestamptz			now()	

Constraint, index, or trigger	Definition
symptom_reports_severity_check	CHECK (((severity >= 0) AND (severity <= 10)))
symptoms_other_ck	CHECK (((symptom <> 'other') OR (btrim(COALESCE(description, '')) <> '')))
symptom_reports_audit_trg	trigger function audit_phi_change()

appointments

27 sample rows · row-level security

Intake, weekly check-in, graduation, and ad hoc visits.

Column	Type	Null	Key	Default	Notes
appointment_id	uuid		**PK**	gen_random_uuid()	
patient_id	uuid		**FK patients.patient_id**		
clinician_id	uuid		**FK users.user_id**		
appointment_type	appointment_type				values: intake, weekly_checkin, graduation_assessment, ad_hoc
modality	appointment_modality			'telehealth_video'	values: telehealth_video, phone, in_person
status	appointment_status			'scheduled'	values: scheduled, completed, missed, cancelled
scheduled_start	timestamptz				
scheduled_end	timestamptz				
cancel_reason	varchar(255)	yes			
notes	text	yes			
created_by	uuid	yes	**FK users.user_id**		
created_at	timestamptz			now()	
updated_at	timestamptz			now()	

Constraint, index, or trigger	Definition
appt_cancel_ck	CHECK (((status <> 'cancelled') OR (cancel_reason IS NOT NULL)))
appt_length_ck	CHECK ((((scheduled_end - scheduled_start) >= '00:10:00') AND ((scheduled_end - scheduled_start) <= '03:00:00')))

appt_clinician_no_overlap	EXCLUDE USING gist (clinician_id WITH =, tstzrange(scheduled_start, scheduled_end) WITH &&) WHERE ((status = ANY (ARRAY['scheduled', 'completed'])))
appt_patient_no_overlap	EXCLUDE USING gist (patient_id WITH =, tstzrange(scheduled_start, scheduled_end) WITH &&) WHERE ((status = ANY (ARRAY['scheduled', 'completed'])))
appt_one_graduation_ux	UNIQUE INDEX appt_one_graduation_ux (patient_id) WHERE ((appointment_type = 'graduation_assessment') AND (status <> 'cancelled'))
appointments_audit_trg	trigger function audit_phi_change()
appt_roles_trg	trigger function validate_roles()

alert_rules 12 sample rows

Program-default threshold, pattern, symptom, and data-gap rules.

Column	Type	Null	Key	Default	Notes
rule_id	uuid		**PK**	gen_random_uuid()	
rule_code	varchar(40)		**UQ**		
name	varchar(120)				
description	text				
rule_type	alert_rule_type				values: threshold, pattern, symptom, data_gap
metric	varchar(40)				
comparator	varchar(2)				
default_threshold	numeric(8,2)				
applies_to_context	reading_context[]	yes			values: rest, warmup, exercise, cooldown, recovery
evaluation_window_minutes	integer			0	
consecutive_required	smallint			1	
severity	alert_severity				values: critical, high, moderate, low
suppression_minutes	integer			60	
is_customizable	boolean			true	
is_active	boolean			true	
updated_by	uuid	yes	**FK users.user_id**		
created_at	timestamptz			now()	
updated_at	timestamptz			now()	

Constraint, index, or trigger	Definition

alert_rules_comparator_check	CHECK (((comparator) = ANY ((ARRAY['>', '>=', '<', '<=', '=']))))
alert_rules_consecutive_required_check	CHECK (((consecutive_required >= 1) AND (consecutive_required <= 20)))
alert_rules_evaluation_window_minutes_check	CHECK (((evaluation_window_minutes >= 0) AND (evaluation_window_minutes <= 43200)))
alert_rules_rule_code_check	CHECK (((rule_code) ~ '^[A-Z][A-Z0-9_]{2,39}$'))
alert_rules_suppression_minutes_check	CHECK (((suppression_minutes >= 0) AND (suppression_minutes <= 10080)))
rules_critical_ck	CHECK (((severity <> 'critical') OR (suppression_minutes = 0)))
alert_rules_rule_code_key	UNIQUE (rule_code)

patient_alert_thresholds

2 sample rows · row-level security

Physician-set patient-specific threshold overrides.

Column	Type	Null	Key	Default	Notes
override_id	uuid		**PK**	gen_random_uuid()	
patient_id	uuid		**FK patients.patient_id**		
rule_id	uuid		**FK alert_rules.rule_id**		
threshold_value	numeric(8,2)				
reason	text				
set_by	uuid		**FK users.user_id**		
effective_from	timestamptz			now()	
effective_to	timestamptz	yes			
created_at	timestamptz			now()	

Constraint, index, or trigger	Definition
patient_alert_thresholds_reason_check	CHECK ((length(btrim(reason)) >= 5))
thresholds_period_ck	CHECK (((effective_to IS NULL) OR (effective_to > effective_from)))
thresholds_one_current_ux	UNIQUE INDEX thresholds_one_current_ux (patient_id, rule_id) WHERE (effective_to IS NULL)
patient_alert_thresholds_audit_trg	trigger function audit_phi_change()
thresholds_check_trg	trigger function threshold_override_check()
thresholds_roles_trg	trigger function validate_roles()

escalation_policies

4 sample rows

Acknowledgment timeouts, escalation targets, and channels by severity.

Column	Type	Null	Key	Default	Notes
severity	alert_severity		**PK**		values: critical, high, moderate, low
ack_timeout_minutes	integer				
escalate_to	care_team_role				values: primary_clinician, supervising_physician, covering_clinician
max_escalation_level	smallint			2	
channels	notification_channel[]				values: dashboard, email, sms
deliver_as_digest	boolean			false	
updated_by	uuid	yes	**FK users.user_id**		
updated_at	timestamptz			now()	

Constraint, index, or trigger	Definition
escalation_critical_ck	CHECK (((severity <> 'critical') OR ((ack_timeout_minutes <= 30) AND (NOT deliver_as_digest))))
escalation_digest_ck	CHECK (((NOT deliver_as_digest) OR (severity = 'low')))
escalation_policies_ack_timeout_minutes_check	CHECK (((ack_timeout_minutes >= 1) AND (ack_timeout_minutes <= 1440)))
escalation_policies_channels_check	CHECK ((cardinality(channels) >= 1))
escalation_policies_max_escalation_level_check	CHECK (((max_escalation_level >= 0) AND (max_escalation_level <= 3)))

alerts 6 sample rows row-level security

Raised alerts with full acknowledgment, escalation, and resolution lifecycle.

Column	Type	Null	Key	Default	Notes
alert_id	uuid		**PK**	gen_random_uuid()	
patient_id	uuid		**FK patients.patient_id**		
rule_id	uuid		**FK alert_rules.rule_id**		
vital_id	bigint	yes	**FK vital_signs.vital_id**		
session_id	uuid	yes	**FK exercise_sessions.session_id**		
symptom_id	uuid	yes	**FK symptom_reports.symptom_id**		
severity	alert_severity				values: critical, high, moderate, low

status	alert_status			'open'	values: open, acknowledged, escalated, resolved, suppressed
title	varchar(160)				
triggering_values	jsonb			'{}'	
threshold_applied	numeric(8,2)	yes			
dedup_key	varchar(120)				Identifies the same underlying condition so repeats attach to one open alert instead of creating new ones.
triggered_at	timestamptz			now()	
acknowledged_at	timestamptz	yes			
acknowledged_by	uuid	yes	**FK users.user_id**		
escalation_level	smallint			0	
escalated_at	timestamptz	yes			
resolved_at	timestamptz	yes			
resolved_by	uuid	yes	**FK users.user_id**		
resolution_note	text	yes			

Constraint, index, or trigger	Definition
alerts_ack_ck	CHECK ((((acknowledged_at IS NULL) = (acknowledged_by IS NULL)) AND ((acknowledged_at IS NULL) OR (acknowledged_at >= triggered_at))))
alerts_ack_state_ck	CHECK (((status <> 'acknowledged') OR (acknowledged_at IS NOT NULL)))
alerts_escalate_ck	CHECK (((status <> 'escalated') OR ((escalation_level >= 1) AND (escalated_at IS NOT NULL))))
alerts_escalation_level_check	CHECK (((escalation_level >= 0) AND (escalation_level <= 3)))
alerts_resolved_ck	CHECK ((((status = 'resolved') = (resolved_at IS NOT NULL)) AND ((status <> 'resolved') OR ((resolved_by IS NOT NULL) AND (btrim(COALESCE(resolution_note, '')) <> '') AND (resolved_at >= triggered_at)))))
alerts_triggering_values_check	CHECK ((jsonb_typeof(triggering_values) = 'object'))
alerts_open_dedup_ux	UNIQUE INDEX alerts_open_dedup_ux (patient_id, rule_id, dedup_key) WHERE (status = ANY (ARRAY['open', 'acknowledged', 'escalated']))
alerts_audit_trg	trigger function audit_phi_change()
alerts_history_trg	trigger function alert_history()
alerts_roles_trg	trigger function validate_roles()

alert_events 13 sample rows row-level security

Append-only alert history written automatically by trigger.

Column	Type	Null	Key	Default	Notes
event_id	bigint		**PK**	identity	
alert_id	uuid		**FK alerts.alert_id**		
event_type	alert_event_type				values: created, notified, acknowledged, escalated, resolved, suppressed, comment
from_status	alert_status	yes			values: open, acknowledged, escalated, resolved, suppressed
to_status	alert_status	yes			values: open, acknowledged, escalated, resolved, suppressed
actor_user_id	uuid	yes	**FK users.user_id**		
occurred_at	timestamptz			now()	
note	text	yes			
payload	jsonb			'{}'	

Constraint, index, or trigger	Definition
alert_events_payload_check	CHECK ((jsonb_typeof(payload) = 'object'))
alert_events_append_only	trigger function forbid_change()

notifications 8 sample rows row-level security

Per-recipient, per-channel delivery records with a documented payload.

Column	Type	Null	Key	Default	Notes
notification_id	uuid		**PK**	gen_random_uuid()	
alert_id	uuid		**FK alerts.alert_id**		
recipient_user_id	uuid		**FK users.user_id**		
channel	notification_channel				values: dashboard, email, sms
status	notification_status			'queued'	values: queued, sent, delivered, failed
payload	jsonb				
attempts	smallint			0	
queued_at	timestamptz			now()	

sent_at	timestamptz	yes
delivered_at	timestamptz	yes
read_at	timestamptz	yes
failure_reason	varchar(255)	yes

Constraint, index, or trigger	Definition
notif_failed_ck	CHECK (((status <> 'failed') OR (failure_reason IS NOT NULL)))
notif_order_ck	CHECK (((delivered_at IS NULL) OR (delivered_at >= sent_at)))
notif_sent_ck	CHECK (((status <> ALL (ARRAY['sent', 'delivered'])) OR (sent_at IS NOT NULL)))
notifications_attempts_check	CHECK (((attempts >= 0) AND (attempts <= 10)))
notifications_payload_check	CHECK ((payload ?& ARRAY['alert_id', 'patient_id', 'severity', 'rule_code', 'triggered_at']))
notifications_pending_ix	INDEX notifications_pending_ix (queued_at) WHERE (status = 'queued')

risk_level_bands 3 sample rows

Configurable score boundaries and display colors for risk levels.

Column	Type	Null	Key	Default	Notes
level	risk_level		**PK**		values: low, moderate, high
min_score	numeric(5,2)				
max_score	numeric(5,2)				
color_hex	char(7)				
label	varchar(40)				

Constraint, index, or trigger	Definition
bands_order_ck	CHECK ((min_score < max_score))
risk_level_bands_color_hex_check	CHECK ((color_hex ~ '^#[0-9A-Fa-f]{6}$'))
risk_level_bands_max_score_check	CHECK (((max_score >= (0)) AND (max_score <= (100))))
risk_level_bands_min_score_check	CHECK (((min_score >= (0)) AND (min_score <= (100))))
bands_no_overlap	EXCLUDE USING gist (numrange(min_score, max_score, '[)') WITH &&)

risk_assessments 15 sample rows row-level security

Model output per patient per run: score, level, probabilities.

Column	Type	Null	Key	Default	Notes

assessment_id	uuid		**PK**	gen_random_uuid()	
patient_id	uuid		**FK** **patients.patient_id**		
assessed_at	timestamptz			now()	
model_name	varchar(60)				
model_version	varchar(20)				
window_start	timestamptz				
window_end	timestamptz				
risk_score	numeric(5,2)				
risk_level	risk_level				values: low, moderate, high
dropout_probability	numeric(4,3)	yes			
decline_probability	numeric(4,3)	yes			
created_at	timestamptz			now()	

Constraint, index, or trigger	Definition
risk_assessments_decline_probability_check	CHECK (((decline_probability >= (0)) AND (decline_probability <= (1))))
risk_assessments_dropout_probability_check	CHECK (((dropout_probability >= (0)) AND (dropout_probability <= (1))))
risk_assessments_model_version_check	CHECK (((model_version) ~ '^[0-9]+\.[0-9]+\.[0-9]+$'))
risk_assessments_risk_score_check	CHECK (((risk_score >= (0)) AND (risk_score <= (100))))
risk_window_ck	CHECK (((window_end > window_start) AND (window_end <= (assessed_at + '00:01:00'))))
risk_level_trg	trigger function risk_level_from_band()

risk_factors

10 sample rows · row-level security

Ranked contributing factors that explain each assessment.

Column	Type	Null	Key	Default	Notes
factor_id	uuid		**PK**	gen_random_uuid()	
assessment_id	uuid		**FK** **risk_assessments.assessment_id**		
factor_code	varchar(40)				
label	varchar(120)				
observed_value	varchar(60)				
reference_value	varchar(60)	yes			
contribution	numeric(6,3)				

rank	smallint

Constraint, index, or trigger	Definition
risk_factors_factor_code_check	CHECK (((factor_code) ~ '^[a-z][a-z0-9_]{2,39}$'))
risk_factors_rank_check	CHECK (((rank >= 1) AND (rank <= 20)))
factors_code_uq	UNIQUE (assessment_id, factor_code)
factors_rank_uq	UNIQUE (assessment_id, rank)

care_recommendations

4 sample rows · row-level security

System-generated suggestions that require clinician review.

Column	Type	Null	Key	Default	Notes
recommendation_id	uuid		**PK**	gen_random_uuid()	
assessment_id	uuid	yes	**FK risk_assessments.assessment_id**		
patient_id	uuid		**FK patients.patient_id**		
recommendation_text	text				
source	varchar(20)			'system'	
status	recommendation_status			'pending_review'	values: pending_review, approved, rejected
reviewed_by	uuid	yes	**FK users.user_id**		
reviewed_at	timestamptz	yes			
review_note	text	yes			
created_at	timestamptz			now()	

Constraint, index, or trigger	Definition
care_recommendations_recommendation_text_check	CHECK ((length(btrim(recommendation_text)) >= 10))
care_recommendations_source_check	CHECK (((source) = ANY ((ARRAY['system', 'clinician']))))
rec_review_ck	CHECK ((((status = 'pending_review') = (reviewed_by IS NULL)) AND ((reviewed_by IS NULL) = (reviewed_at IS NULL))))
rec_system_link_ck	CHECK ((((source) <> 'system') OR (assessment_id IS NOT NULL)))
rec_pending_ix	INDEX rec_pending_ix (created_at) WHERE (status = 'pending_review')
care_recommendations_audit_trg	trigger function audit_phi_change()
rec_roles_trg	trigger function validate_roles()

audit_log 120 sample rows row-level security

Append-only HIPAA audit trail; stores who touched which record, not PHI values.

Column	Type	Null	Key	Default	Notes
audit_id	bigint		PK	identity	
occurred_at	timestamptz			now()	
actor_user_id	uuid	yes			
actor_role	user_role	yes			values: patient, clinician, physician, admin, it_admin
action	audit_action				values: create, read, update, delete, login, login_failed, logout, export, acknowledge
entity_type	varchar(40)				
entity_id	text	yes			
patient_id	uuid	yes			
outcome	varchar(10)			'success'	
request_id	uuid	yes			
ip_address	inet	yes			
details	jsonb			'{}'	

Constraint, index, or trigger	Definition
audit_log_details_check	CHECK ((jsonb_typeof(details) = 'object'))
audit_log_outcome_check	CHECK (((outcome) = ANY ((ARRAY['success', 'denied', 'error']))))
audit_patient_ix	INDEX audit_patient_ix (patient_id, occurred_at DESC) WHERE (patient_id IS NOT NULL)
audit_log_append_only	trigger function forbid_change()
audit_log_no_truncate	trigger function forbid_change()

Enumerated types

Type	Allowed values (in sort order)
alert_event_type	created, notified, acknowledged, escalated, resolved, suppressed, comment
alert_rule_type	threshold, pattern, symptom, data_gap
alert_severity	critical, high, moderate, low
alert_status	open, acknowledged, escalated, resolved, suppressed
appointment_modality	telehealth_video, phone, in_person
appointment_status	scheduled, completed, missed, cancelled
appointment_type	intake, weekly_checkin, graduation_assessment, ad_hoc
audit_action	create, read, update, delete, login, login_failed, logout, export, acknowledge
care_team_role	primary_clinician, supervising_physician, covering_clinician
clinical_risk	low, moderate, high
data_quality	valid, suspect, artifact
device_status	active, inactive, lost, retired
device_type	heart_rate_band, blood_pressure_cuff, pulse_oximeter, activity_tracker
notification_channel	dashboard, email, sms
notification_status	queued, sent, delivered, failed
program_status	referred, active, paused, graduated, withdrawn
qualifying_event	myocardial_infarction, pci, cabg, valve_surgery, stable_angina, heart_failure_reduced_ef, heart_transplant
reading_context	rest, warmup, exercise, cooldown, recovery
reading_source	device, manual
recommendation_status	pending_review, approved, rejected
risk_level	low, moderate, high
session_status	in_progress, completed, partial, aborted
session_type	aerobic, resistance, combined, flexibility
sex_at_birth	female, male, intersex, unknown
symptom_type	chest_pain, shortness_of_breath, dizziness, palpitations, unusual_fatigue, leg_pain, nausea, other
user_role	patient, clinician, physician, admin, it_admin

Views

View	Purpose

v_alert_response	Minutes to acknowledge and resolve per alert, for quality review and threshold tuning. Security invoker.
v_daily_vital_summary	Daily min, max, and average vitals per context in the patient's time zone; feeds trend charts and model features. Security invoker.
v_patient_overview	Clinician worklist: current prescription, care team, sessions done, last reading, open alerts, latest risk. Security invoker, so row-level security applies.
v_program_metrics	Aggregate counts and rates with no identifiers; the administrator's view of the program. Runs with owner rights by design.
v_weekly_adherence	Sessions completed versus prescribed per ISO week, minutes in zone, average exercise HR and RPE. Security invoker.